

Ex No: 02	Scheduling of Project with PERT and CPM Techniques
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Aim

To estimate the project completion time by creating a project schedule, defining task dependencies, generating the CPM and PERT network diagrams, assigning resources and costs, and calculating the overall project duration using ProjectLibre

Algorithm

1. Open ProjectLibre.
2. Create a new project.
3. Enter project name and start date.
4. Create project activities with durations.
5. Define predecessor relationships.
6. Generate Gantt Chart.
7. Generate Network Diagram (PERT/CPM).
8. Create Resource Sheet.
9. Assign resource costs.
10. Allocate resources.
11. View Project Summary and Cost Report.
12. Save the project

Procedure

Step 1: Create a New Project

1. Open **ProjectLibre**.
2. Click **File** → **New**.
3. Enter:
 - **Project Name:** Software Project Scheduling
 - **Manager:** Your Name
 - **Start Date:** Select today's date.

- Click **OK**.

Step 2: Enter Project Activities

In the **Gantt Chart**, enter the following activities:

Task Name	Duration
Requirement Gathering	2d
Feasibility Analysis	3d
UI/UX Design	4d
Database Development	5d
Application Development	6d
System Testing	3d
Bug Resolution	2d
Project Deployment	1d

Step 3: Enter Predecessors

In the **Predecessors** column, enter:

Task	Predecessor
Requirement Gathering	-
Feasibility Analysis	1
UI/UX Design	2
Database Development	3

Task	Predecessor
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Application Development	4
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System Testing	5
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Bug Resolution	6
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Project Deployment	7
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ProjectLibre will automatically schedule the project.

Step 4: View the Gantt Chart

After entering the tasks and predecessors:

- The **Gantt Chart** will be generated automatically on the right side.
- This shows the start date, finish date, and duration of each activity.

Step 5: View the Network Diagram (PERT/CPM)

1. Click **View**.
2. Select **Network Diagram**.

ProjectLibre automatically creates the **PERT/CPM Network Diagram** showing the sequence of project activities.

Step 6: Create Resources

Go to **View** → **Resource Sheet** and enter:

Resource Name	Standard Rate
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Arun Kumar	₹2000/day
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Resource Name Standard Rate

Priya Sharma	₹1800/day
Karthik Raj	₹2500/day
Nisha Devi	₹2200/day
Rahul Verma	₹2700/day
Deepika S	₹2100/day
Ajay Kumar	₹1900/day
Meena R	₹2300/day

Step 7: Assign Resources

Return to the **Gantt Chart**.

Assign the resources:

Task	Resource
Requirement Gathering	Arun Kumar
Feasibility Analysis	Priya Sharma
UI/UX Design	Karthik Raj
Database Development	Nisha Devi
Application Development	Rahul Verma
System Testing	Deepika S
Project Deployment	Meena R
Bug Resolution	Ajay Kumar

Step 8: View Project Summary

After assigning resources:

- Click **Project** or **Reports** (depending on your ProjectLibre version).
- View:
 - Total Duration
 - Total Cost
 - Resource Allocation

Observation:

Project Duration: 26 Days 208 Hours

Critical Path:

Requirement Gathering → Feasibility Analysis → UI/UX Design → Database Development
→ Application Development → System Testing → Bug Resolution → Project Deployment

[illegible]

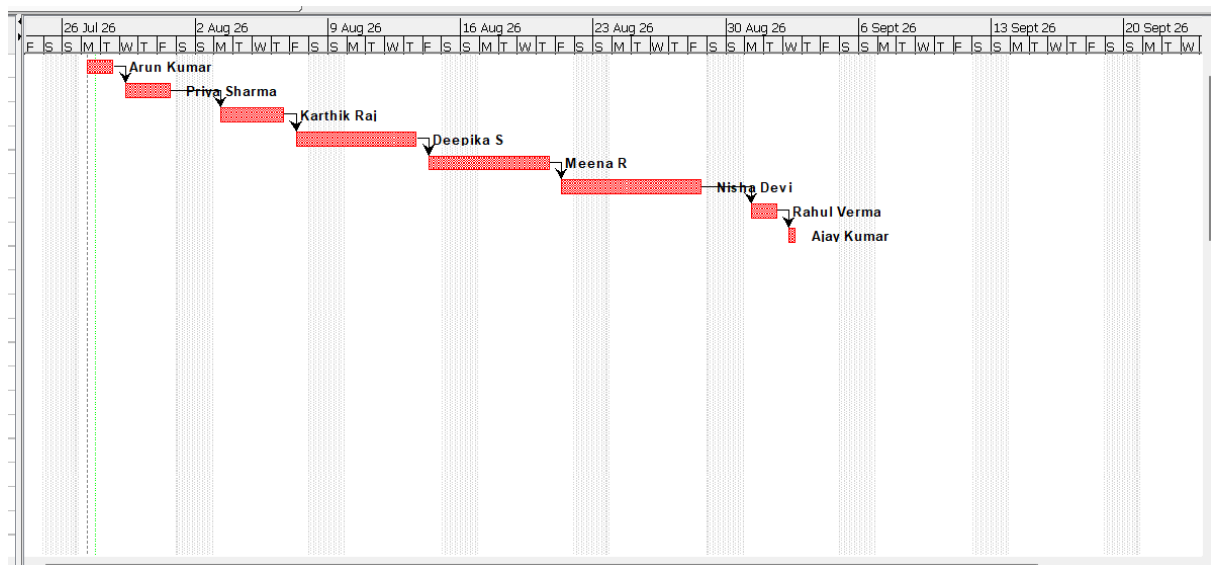
cost estimation *

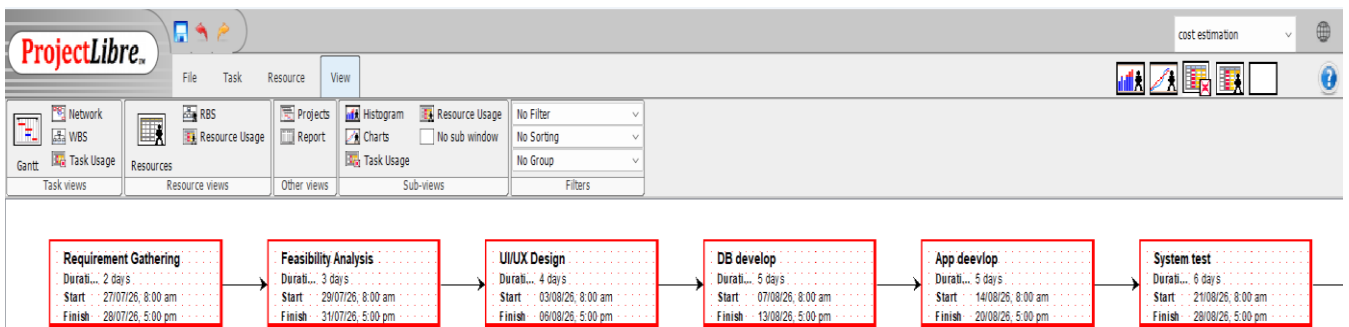
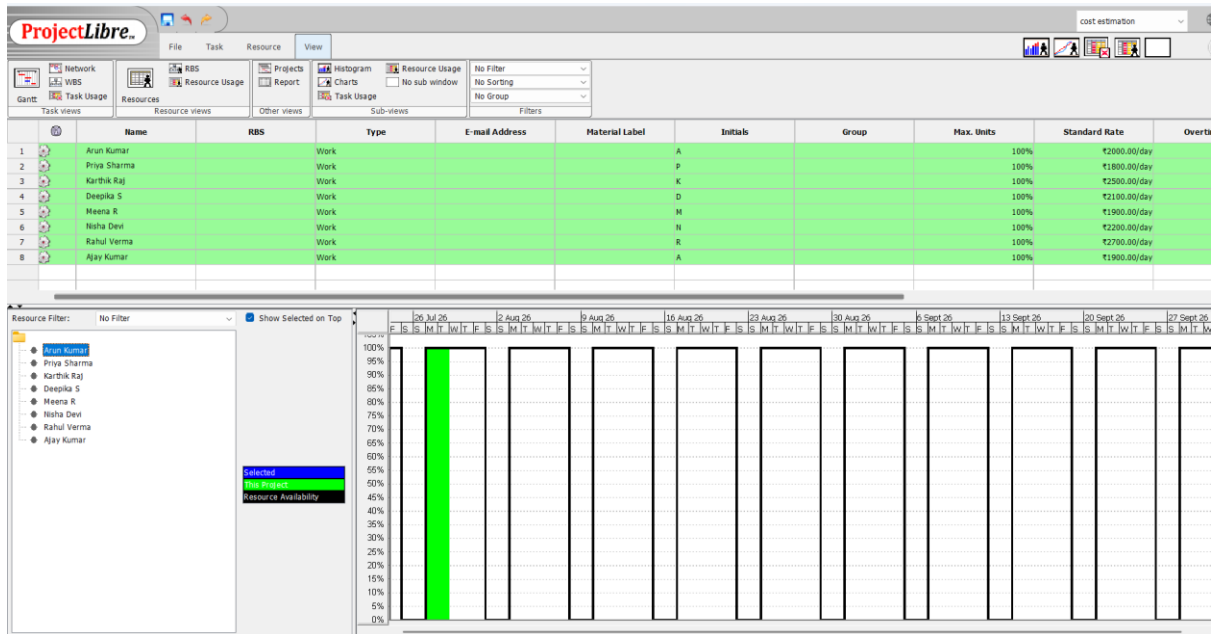
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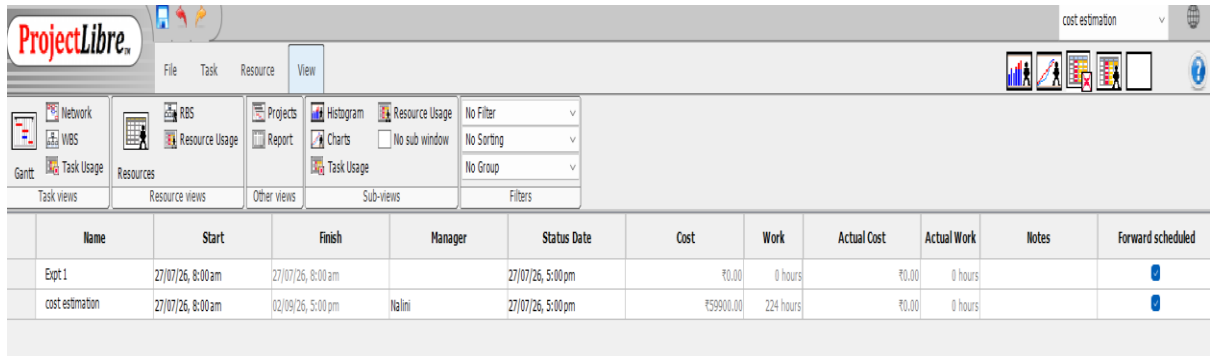
File Task Resource View

Network WBS Task Usage Resources Resource Usage Projects Report Histogram Charts Resource Usage No Filter No sub window No Sorting No Group Task Usage Sub-views Filters

	Name	RBS	Type	E-mail Address	Material Label	Initials	Group	Max. Units	Standard Rate	Overtime
1	Arun Kumar		Work		A			100%	₹2000.00/day	
2	Priva Sharma		Work		P			100%	₹1800.00/day	
3	Karthik Raj		Work		K			100%	₹2500.00/day	
4	Deepika S		Work		D			100%	₹2100.00/day	
5	Meena R		Work		M			100%	₹1900.00/day	
6	Nisha Devi		Work		N			100%	₹2200.00/day	
7	Rahul Verma		Work		R			100%	₹2700.00/day	
8	Ajay Kumar		Work		A			100%	₹1900.00/day	







The screenshot shows the ProjectLibre software interface. The top menu bar includes File, Task, Resource, and View. The View menu is open, showing options like Network, RBS, Projects, Histogram, Resource Usage, No Filter, No Sorting, No Group, and Task Usage. Below the menu bar, there are tabs for Task views, Resource views, Other views, and Sub-views. The main area displays a table with the following data:

Name	Start	Finish	Manager	Status Date	Cost	Work	Actual Cost	Actual Work	Notes	Forward scheduled
Expt 1	27/07/26, 8:00 am	27/07/26, 8:00 am		27/07/26, 5:00 pm	₹0.00	0 hours	₹0.00	0 hours		
cost estimation	27/07/26, 8:00 am	02/09/26, 5:00 pm	Nalini	27/07/26, 5:00 pm	₹59900.00	224 hours	₹0.00	0 hours		

Result:

The project schedule was successfully created using **PERT and CPM techniques** in ProjectLibre. The software automatically generated the **Gantt Chart** and **Network Diagram**, identified the **critical path**, and estimated the **project completion time of 26 days (208 hours)**.